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Lab 1: Requirements Table

Problem Statement 53 — Freelance Content Creator Escrow Platform

Actors: Content Creator, Client Sponsor (Payment Gateway as external supporting actor)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow creators to submit draft media deliverables for client review and release locked milestone payments upon sponsor approval sign-off.	High	Pass: Milestone marked completed and funds credited to creator balance. Fail: Funds released without client sign-off.	Core escrow function protecting both creator payment and sponsor deliverable quality.
FR-002	Functional	The system shall allow a client sponsor to review a watermarked draft asset and either approve it or reject it with written comments.	High	Pass: Sponsor's approve/reject action updates the milestone status and comments are stored and visible to the creator. Fail: Milestone status unchanged after sponsor action.	Enables the sponsor sign-off step required before payment release.
FR-003	Functional	The system shall allow a creator and sponsor to jointly define milestone deliverables, due dates, and payment amounts when creating a contract.	High	Pass: A saved contract shows at least one milestone with a due date and payment amount linked to it. Fail: Contract can be saved with an undefined milestone.	Milestones are the basis for tracking progress and triggering payments.
FR-004	Functional	The system shall automatically apply a visible watermark to any uploaded draft media file before it is made available to the client sponsor for review.	Medium	Pass: Every draft file viewed by the sponsor displays the watermark overlay. Fail: An unwatermarked draft is viewable by the sponsor.	Protects creator's unpaid work from unauthorized use prior to payment.
FR-005	Functional	The system shall send a notification to both the creator and the sponsor whenever a milestone's status changes (submitted, approved, rejected, or paid).	Medium	Pass: Both parties receive an in-app or email notification within 1 minute of a status change. Fail: A status change occurs with no notification sent.	Keeps both stakeholders informed and reduces missed deadlines or disputes.
NFR-001	Performance & Security	Uploaded media drafts must process automated watermarking in under 5 seconds for video/image previews.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Keeps the review workflow responsive for creators and sponsors.
NFR-002	Security	The system shall encrypt all escrow fund transfers and payment transactions in transit (TLS 1.2 or higher) and maintain an immutable audit log of every milestone payment release.	High	Pass: Security scan confirms TLS 1.2+ on all payment endpoints and every payment release event has a corresponding, unalterable audit log entry. Fail: Any payment endpoint uses an unencrypted channel or a release has no log entry.	Protects sensitive financial data and provides traceability for dispute resolution.

Use-Case Flow Specification

Use Case: UC-08 Release Milestone Payment

Primary Actors: Client Sponsor, Content Creator

Supporting Actor: Payment Gateway

Preconditions

1. A contract with at least one defined milestone exists between the creator and sponsor.
2. The creator has submitted the draft deliverable for that milestone (UC-03), and it has been watermarked (UC-04).
3. The sponsor has reviewed the draft (UC-05) and approved it (UC-07).
4. Escrow funds for the milestone are held in the platform's escrow account.

Postconditions

- Success: The milestone status is set to "Paid", escrow funds are transferred to the creator's balance, and both parties are notified.
- Failure: The milestone status remains "Approved" (not "Paid"), no funds move, and the sponsor is notified of the payment failure.

Main Success Scenario

1. The sponsor confirms milestone approval and selects "Release Payment" for the completed milestone.
2. The system verifies that the milestone status is "Approved" and that sufficient escrow funds are held.
3. The system invokes the included use case UC-09 Process Payment, connecting to the Payment Gateway to authorize transfer of the milestone amount.
4. The Payment Gateway confirms the transaction and returns an authorization reference.
5. The system updates the milestone status to "Paid" and credits the amount to the creator's balance.
6. The system triggers UC-10 Receive Notification, informing both the creator and sponsor that payment has been released.
7. The system logs the transaction (amount, timestamp, authorization reference) to the audit log. Use case ends successfully.

Alternate Flow — 4a. Payment Authorization Declined

- 4a1. The Payment Gateway returns a decline response instead of an authorization.
- 4a2. The system keeps the milestone status as "Approved" and does not move any escrow funds.
- 4a3. The system notifies the sponsor that the payment attempt failed and prompts them to retry or update the payout method.
- 4a4. If the payment fails twice, the milestone is flagged for manual review by platform support, and the creator is notified of the delay. Use case ends without payment.

Use-Case Diagram

